

Grade 7 Accelerated
Unit 1
Lesson 9 Practice

Name _____
Date _____
Period _____

For questions 1–4, use the inequality $3(7 - 2h) > 15$.

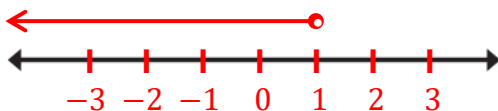
1. Solve $3(7 - 2h) > 15$ by using the distributive property first.

$$\begin{array}{rcl} 21 - 6h & > & 15 \\ -21 & -21 & \\ \hline -6h & > & -6 \\ \hline -6 & > & -6 \end{array} \quad h < 1$$

2. Solve $\frac{3(7 - 2h)}{3} > \frac{15}{3}$ by eliminating the 3 first.

$$\begin{array}{rcl} 7 - 2h & > & 5 \\ -7 & -7 & \\ \hline -2h & > & -2 \\ \hline -2 & > & -2 \end{array} \quad h < 1$$

3. Graph the solution to the inequality.

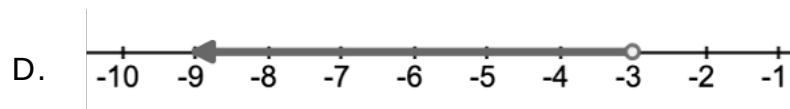
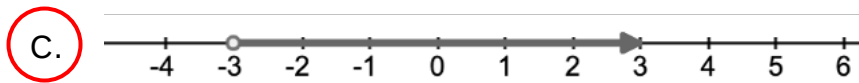
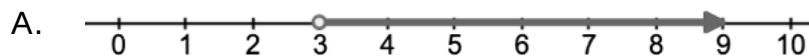


4. Use substitution to verify the solution set.

$$\begin{array}{l} 3(7 - 2 \cdot 0) > 15 \\ 3(7) > 15 \\ 21 > 15 \checkmark \end{array} \quad \text{When } h < 1, \text{ zero (or any number less than 1 makes a true statement)}$$

5. Which of the following represents the solution to the inequality below?

$$-8 < \frac{2}{3}(c - 9) \quad -8 \leq \frac{2}{3}(c - 9) \quad \rightarrow \quad \frac{3}{2} \cdot -8 \leq \frac{3}{2} \cdot \frac{2}{3}(c - 9)$$



$$\frac{-24}{2} \leq c - 9$$

$$-12 \leq c - 9$$

$$+9 \quad +9$$

$$-3 \leq c$$

$$c \geq -3$$

For questions 6–9, solve each inequality and graph the solution set using whichever method you prefer.

6. $6(g - 2) \leq 18$

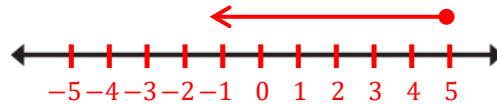
Use distributive property or division. Division shown here:

$$\frac{6(g - 2)}{6} \leq \frac{18}{6}$$

$$g - 2 \leq 3$$

$$+2 \quad +2$$

$$g \leq 5$$



7. $21.6 \leq 7.2(12 + x)$

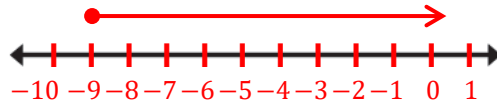
Use distributive property or division. Distributive property shown here.

$$21.6 \leq 86.4 + 7.2x$$

$$-86.4 \quad -86.4$$

$$\frac{-64.8}{7.2} \leq \frac{7.2x}{7.2}$$

$$-9 \leq x$$



8. $-\frac{5}{2}\left(\frac{3}{4}y + 6\right) < -22\frac{1}{2}$

Use distributive property or division. Distributive shown here.

$$\frac{-15}{8}y + \frac{-30}{2} < -22\frac{1}{2}$$

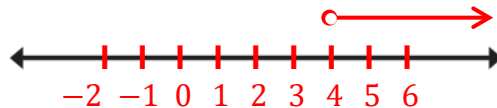
$$\frac{-8}{15} \cdot \frac{-15}{8}y < \frac{-15}{2} \cdot \frac{-8}{15}$$

$$\frac{-15}{8}y + \frac{-30}{2} < -\frac{45}{2}$$

$$y > \frac{8}{2}$$

$$\frac{+30}{2} \quad \frac{+30}{2}$$

$$y > 4$$



9. $\frac{3}{8}(p + 29) > 9$

Use distributive property or division. Division shown here.

$$\frac{8}{3} \cdot \frac{3}{8}(p + 29) > 9 \cdot \frac{8}{3}$$

$$p + 29 > \frac{72}{3}$$

$$p + 29 > 24$$

$$-29 \quad -29$$

$$p > -5$$



10. Your friend was absent from class today. Describe how to solve the inequality $-10 \geq -2.5(z + 8)$ in two ways.

1. Using the distributive property:

$$-10 \geq -2.5z - 20$$

$$+20 \quad +20$$

$$\frac{10}{-2.5} \geq \frac{-2.5z}{-2.5}$$

$$-4 \leq z$$

2. Use division:

$$\frac{-10}{-2.5} \geq \frac{-2.5(z + 8)}{-2.5}$$

$$4 \leq z + 8$$

$$-8 \quad -8$$

$$-4 \leq z$$